Experiment No. 13

Title : To study the properties of baking soda and washing soda.

Apparatus : Test tubes, Test tube stand, Petrydishes, White tile etc.

Chemicals : Baking Soda(Sodium bicarbonate- $NaHCO_3$),Washing soda(Sodium carbonate - $Na_2CO_3 .10 H_2O$), Litmus paper, pH paper etc.

Procedure :

1. Take one spoonful each of baking and washing soda in two seperate dishes respectively.
2. Leave the two petrydishes on table. Observe the two samples carefully after an half hour.
3. Prepare aqueous solutions of the two samples.
4. Test the solutions using litmus paper.
5. Place two drops of aqueous solutions of both the samples on pH paper and compare the colour shades.
6. Place pH paper strip on white tile and check the pH of two samples of soda.

Observation :

Sample	Observations in petrydish : Nature	Litmus test	pH of Solution	Inference
Baking Soda	White.	Blue litmus - Blue Red litmus - Turns Blue	8-9	It is basic in nature.
Washing Soda	White.	Blue litmus - Blue Red litmus - Blue.	9-10	It is Basic in nature.

Note : On the basis of your observations and other tests, write the properties of baking soda and washing soda in the above chart.

Inference :

Details Yes/No	Detailed Conclusion
Hygroscopic	Backing soda is the base.
Acidic	because base has no colour.
Basic	Non odur of faulsmell and
	test in bitter.

Multiple choice questions

1. The salt ..... is used in making dhokala, cake.
   A. Na_2CO_3 B. NaOH C. $NaHCO_3$ D. Na_2SO_4
2. ..... is used for making soft water from hard water.
   A. NaCl B. NaOH C. $Na_2CO_3.10H_2O$ D. $NaHCO_3$
3. The foodstuff become light and spongy by baking soda because of.....
   A. air is released B. carbon dioxide is released
   C. baking soda is a base D. baking soda is neutral
4. pH of basic salts is.....
   A. 1 to 6 B. 7 C. 1 to 14 D. 8 to 14
5. .....salts are formed by combination of strong acid and weak base.
   A. Acidic B. Basic C. Neutral D. Different than these
6. .....salts are formed by combination of strong base and weak acid.
   A. Acidic B. Neutral C. Basic D. all of these
7. .....salts are formed by combination of strong acid and strong base.
   A. Acidic B. Neutral C. Basic D. all of these
8. Identify the odd one out
   A. Potassium sulphate B. Sodium chloride C. Sodium acetate D. Potassium chloride

Questions

1. Explain the differences in baking soda and washing soda.

Backing Soda	washing Soda
1) Backing soda is white amaraphous powder.	1) sodium carbonate is a white crustline substance.
2) Backing soda is a to pressure lights.	2) It is used as cleaning agent in washing soda.

2. Why some kind of detergent powders become moist in rainy season?
   The top water has a typical strong colours in rainy season hence deary ent powder become moist in raing season.

3. Why is the baking soda used in the preparation of cake, biscuit, dhokala? What does happen exactly?
   While making a cake it all mathers odder Backing in is used to Be case acidity in the Stomach.

VSFTUU